American Computer Science League

2025-2026 ● Contest 2: Short Problems ● Intermediate Division

1. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if a = 3, b = -2, and c = 1:

a b * c + a 2 ^ − b c / +

A. -16
B. -14
C. 0
D. 14
E. 18

2. Prefix/Infix/Postfix Notation

Translate the following infix expression into prefix:

$$\frac{y^2z - z^2y}{y + z}$$

A. / − * ^ 2 y z * ^ 2 z y + z y
B. / − ^ y 2 z * * ^ z 2 y + y z
C. − / * ^ y 2 z * ^ z 2 y + y z
D. / − * ^ y 2 z * ^ z 2 y + y z
E. / − * ^ * ^ + y 2 z z 2 y y z

3. Bit-String Flicking

Evaluate:

(LCIRC-2 (RSHIFT-1 (NOT 01110))) OR (RCIRC-2 10011)

A. 11011
B. 11110
C. 11101
D. 10111
E. 01111

4. Bit-String Flicking

Solve the following equation for X if X is a 5-bit string:

(RCIRC-1 (LSHIFT-2 X AND 01101)) = 00010

A. *1*0*
B. ***01
C. *0*1*
D. ***10
E. 01***

5. LISP Evaluate the following LISP expression: (ADD (MULT 2 3) (SUB 9 3) (DIV 8 4))	A. 12 B. 13 C. 14 D. 15 E. 16
6. LISP Evaluate the following LISP expression: (CDR (CAR (REVERSE (CDR (CAR '((a b (c d)) (a b) c))))))	A. (b (c d)) B. c C. d D. (d) E. (c d)

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1. 前缀/中缀/后缀表示法

当 a = 3, b = -2, 且 c = 1 时，对以下后缀表达式进行求值：

a b * c + a 2 ^ − b c / +

A. -16
B. -14
C. 0
D. 14
E. 18

2. 前缀/中缀/后缀表示法

将以下中缀表达式转换为前缀表达式：

$$\frac{y^2z - z^2y}{y + z}$$

A. / − * ^ 2 y z * ^ 2 z y + z y
B. / − ^ y 2 z * * ^ z 2 y + y z
C. − / * ^ y 2 z * ^ z 2 y + y z
D. / − * ^ y 2 z * ^ z 2 y + y z
E. / − * ^ * ^ + y 2 z z 2 y y z

3. 位运算

计算：

(LCIRC-2 (RSHIFT-1 (NOT 01110))) OR (RCIRC-2 10011)

A. 11011
B. 11110
C. 11101
D. 10111
E. 01111

4. 位运算

已知 X 为一个 5 位二进制串，求解下列方程中的 X：

(RCIRC-1 (LSHIFT-2 X AND 01101)) = 00010

A. *1*0*
B. ***01
C. *0*1*
D. ***10
E. 01***

5. LISP 对下列 LISP 表达式进行求值： (ADD (MULT 2 3) (SUB 9 3) (DIV 8 4))	A. 12 B. 13 C. 14 D. 15 E. 16
6. LISP 对下列 LISP 表达式进行求值： (CDR (CAR (REVERSE (CDR (CAR '((a b (c d)) (a b) c))))))	A. (b (c d)) B. c C. d D. (d) E. (c d)